Rutgers University

Master's Comprehensive Examination

Part I — Theory (90 minutes) — December 3, 1988

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded.
Open book and open notes.

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1. For a random sample of size n from the Poisson distribution,

   (a) Find an unbiased estimator of $\tau(\lambda) = (1 + \lambda)e^{-\lambda}$.

   (b) Find a maximum–likelihood estimator of $\tau(\lambda)$.

   (c) Find the UMVUE of $\tau(\lambda)$.

2. Let the (single) random variable X have density function

$$f(x;\theta) = \begin{cases} \theta(x - \frac{1}{2}) + 1 & \text{if } 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases} \quad \text{and } -2 \le \theta \le 2.$$

   (a) Is there a UMP test of size $\frac{1}{4}$ for H_0: $\theta = 0$ vs H_1: $\theta \neq 0$? (Show.)

   (b) Is there a UMP test of size $\frac{1}{4}$ for H_0: $\theta \le 0$ vs H_1: $\theta > 0$? (Show.)

3. Let X and Y be i.i.d. random variables with a geometric distribution having parameter p in (0,1). If M = min (X,Y) and Z = Y − X, find the joint distribution of M and Z. Are M and Z independent?

4. Let X_i, i = 1,2..., be i.i.d. random variables with $P\{X_i = 1\} = p = 1 - P\{X_i = 0\}$. Let T_k = number of trials required to obtain k successes (ones) and S_n = number of successes in the first n trials. Show that for $k \le r$ and $x \le n$

$$P\{T_k = x \mid S_n = r\} = \frac{\binom{x-1}{k-1}\binom{n-x}{r-k}}{\binom{n}{r}}.$$

5. STATE whether the following are true or false and give a short explanation of your answer in each case.

(a) Let X, Y be independent, positive random variables. Then, $E(X/Y) = E(X)/E(Y)$.

(b) Let X_1, X_2, X_3 be i.i.d. with a continuous probability density function. Then $P(X_1 \leq X_2 \leq X_3) = 1/3$.

(c) If a sufficient statistic exists, then the maximum likelihood estimator is a function of it.

(d) If $X_1,\ldots,X_n$ is a random sample from a normal distribution and M denotes the sample median, then

$$E(M \mid \Sigma X_i) = \bar{X}.$$

p/t3

Rutgers University

Master's Comprehensive Examination

Part II | Applied (90 minutes) | December 3, 1988

Instructions: Attempt any four out of the following five questions. Mention which four you would like to have graded.
Open book and open notes.

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1. A new reducing diet was tested on nine women selected at random, with the results in the table below.

   (a) Give an 80% confidence interval for the mean weight difference in the population represented by the nine women.

   (b) At a 1% significance level, would you conclude that the diet is effective?

Woman	Weight before	Weight after
1	133	132
2	146	143
3	135	137
4	168	160
5	148	151
6	152	148
7	180	171
8	126	120
9	122	124

2. In order to study the efficiency of its staff, a local bank manager decided to observe the traffic in the bank's lobby for an hour. During this period ten customers arrived, of which only seven left before the hour was over. Their individual service times were recorded as 10, 13, 12, 8, 30, 6 and 5 minutes. For the three customers who did not complete their activities at the end of the observation period, the manager recorded only that their individual service times EXCEEDED 3, 7 and 4 minutes.

   Under the assumption that the service times are independent random variables with a common exponential distribution ($F(x) = e^{-\theta x}$, $x > 0$, $\theta > 0$), what should be the manager's estimate of the mean and standard deviation of the service time, if he maximizes the likelihood of the data.

3. An experiment was conducted to test the effect of having coffee, background music, and type of lighting on productivity of workers. There were six treatment combinations. Twelve subjects were distributed completely at random, two subjects to each treatment. The productivity of the workers is the response variate Y. The treatment combinations and the data are as follows:

	Treatment Combination			Y scores	
	Coffee	Lighting	Music		
1	no	cool	no	12,	16
2	no	warm	no	20,	28
3	no	warm	yes	8,	2
4	yes	cool	no	30,	22
5	yes	warm	no	30,	42
6	yes	warm	yes	20,	32

Fill in the following Analysis of Variance Table:

Source	df	SS	MS	F
TREATMENT				
Coffee				
B	2			
Coffee x B				
Error				
Total				

Factor B denotes the combinations of Lighting and Music, and takes three levels (namely: (cool,no); (warm,no); (warm,yes)).

4. Suppose I have the following contingency tables

NEW YORK

	Sick	Well	
Vaccine	20	80	100
NonVaccine	50	50	100
	70	130	

LOS ANGELES

	Sick	Well	
Vaccine	30	70	100
NonVaccine	50	50	100
	80	120	

Test the hypothesis that the vaccine has the same effect in both cities.

5. Ten persons are each driven blindfolded over a predetermined course using two automobiles, A and B. In each case the person rates the quality of the ride on a scale of 0 to 20. The results are as follows:

Person	1	2	3	4	5	6	7	8	9	10
Automobile A	14	20	15	7	16	12	10	13	11	13
Automobile B	10	17	15	9	8	10	9	11	14	8

Do the data suggest that the median difference in score (A − B) over all potential participants is significantly different from zero?

Rutgers University
Master's Comprehensive Re–Exam, 3/26/88

Part II: Applied (90 minutes)

Instructions: Open book and open notes.

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1. An experiment was performed on 26 subjects. Two variables X and Y were measured for subject. The summary values are:

$$\Sigma X_i = 1613 \quad \Sigma Y_i = 281.9 \quad \Sigma X_i Y_i = 16731.0$$
$$\Sigma(X_i - \bar{X})^2 = 3756.96, \quad \Sigma(Y_i - \bar{Y})^2 = 465.34.$$

A simple linear regression $Y = \beta_0 + \beta_1 X + e$ is to be fit to the data.

a) Find the estimated regression line of Y on X.

b) Find a 95% confidence interval for β_1.

c) Find a 90% confidence interval for the mean value of Y when X = 2.

d) Find a 95% confidence interval for ρ.

e) Test the hypothesis $H_0: \rho = -0.5$ vs. $H_A: \rho \le -0.5$ at $\alpha = .1$.

2. The number of major earthquakes occuring in a single year seems to follow the Poisson distribution with parameter λ.
   We have the following data for the number of earthquakes occuring worldwide for the years 1500–1931:

No. of earthquakes (X)	Observed frequency (N_i)
0	223
1	142
2	48
3+	19

Perform a goodness–of–fit test on this data; i.e., test

$$H_0: X \sim \text{Poisson}(\lambda) \quad \text{vs.} \quad H_A: H_0 \text{ is false} \quad \text{at } \alpha = .05.$$

3. A market research study investigated the relationship between an adult's self–perception and his attitude toward a small car. A total of 299 persons were surveyed. Each person was assigned to one of three personality types: 1. conservative, 2. middle–of–the–road, 3. explorer, and asked for an opinion of small cars. The results are below:

	Self–perception 1	2	3
Opinion of small cars			
Favorable	79	58	49
Neutral	10	8	9
Unfavorable	10	34	42

a) Test whether the two traits are independent at $\alpha = .01$.

b) Estimate the probability that a person in the population has a favorable opinion of small cars.

4. The following is a two–way ANOVA table obtained from the data of an experiment on factors A and B.

ANOVA

Source	df.	SS	MS
A	2	4608.17	2304.08
B	1	96.33	96.33
AB interaction	2	283.17	141.58
error	6	139.00	23.17

Suppose that the levels of factor B (but not factor A) were chosen at random from a large population of possible levels.

a) Test all the appropriate hypotheses at $\alpha = .01$.

b) Give simultaneous 99% confidence intervals for all the pairwise differences in the levels of factor A.

The sample means of the three levels of A are:

$$X_{1.} = 55.5, \qquad X_{2.} = 21.25 \qquad X_{3.} = 5.5,$$

each was calculated from four observations.